Applied voltage-induced AgBr formation and associated conduction process changes in $Ag/CH_3NH_3PbBr_3/TiO_2$ heterojunction memory

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Understanding the interaction between the constituent layers of a functional device under an applied field is important to determine the stability of the device and, hence, its applicability. Here, we report the resistive switching behavior of solution-processed $CH_3NH_3PbBr_3/TiO_2$ heterostructure grown on fluorine-doped tin oxide (FTO-coated glass) substrate and studied the interaction of Ag electrode with $CH_3NH_3PbBr_3$ under cyclic voltage sweep operation. The $FTO/TiO_2/CH_3NH_3PbBr_3/Ag$ memristor device shows a low SET and RESET voltage of 0.3 V and −1.4 V, respectively, with an initial ON/OFF ratio of >100. However, with the increase in the cycle number, the formation of AgBr leads to a gradual lowering of the RESET voltage and low resistance state (LRS) current, followed by a stable LRS after about 50 cycles, and the current conduction process in negative bias LRS is changed from filamentary conduction to interface dominated Fowler-Nordheim tunneling conduction. On replacing the Ag electrode with Au, the device shows a stable memristor performance with an ON/OFF ratio of ∼300. Our results show the fundamental limitation of Ag electrodes for $CH_3NH_3PbBr_3$-based resistive switching devices, which may be beneficial in understanding the interaction of electrode materials with other memristor devices.

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I. INTRODUCTION

Considering world digitalization and technological modernization, improving the capacity of data storage memory devices is essential [1,2]. Resistive switching (RS) memory is a type of nonvolatile memory that uses a basic metal-insulator-metal capacitor-like structure, and simply varying the dielectric resistance in the presence of strong electric field results in different resistance states through the development and rupture of conductive filaments, also known as the SET and RESET processes [3–6]. Metal halide perovskites (MHPs) with the general formula of ABX_3 (where A is a cation, B is a divalent metal, and X = Cl, Br, I), a class of semiconducting materials, which have recently attracted significant attraction toward optoelectronics due to their fascinating properties of long diffusion length, tunable band gap, and high absorption coefficient [7–11]. Owing to their low-temperature solution-processed synthesis and easy-to-fabricate device design techniques, MHPs have also been used in transistors and RS memory [12–15]. However, instability in environmental conditions is a significant limitation of MHPs for realization in large-scale applications [16–19]. Hence, understanding the origin of the degradation of constituent layers and their interfaces under the applied field is essential in making a stable MHP-based memristor [20–22].

Most MHP-based switching reports used $CH_3NH_3PbI_3$ as the active material in typical metal-insulator-metal structured RS devices [23–25]. On the contrary, relatively stable cubic structured $CH_3NH_3PbBr_3$ had been used only in a few cases [26–28]. As reported, $CH_3NH_3PbBr_3$-based devices show filamentary switching behavior for thin-layered perovskite devices by conductive filament formation via defective grain boundaries, resulting in device failure after a few sweeping cycles and focused on switching with thicker $CH_3NH_3PbBr_3$ (in μm range) for stable switching performance [28]. However, synthesizing a thick MHP in a solution process is a challenge due to the high solubility of perovskite in the conventionally used solvents like DMF and DMSO, and hence, vapor deposition is used for $CH_3NH_3PbBr_3$-based memory devices [28,29]. To overcome this, we used spin-coated thin-layered bromide perovskite (∼300 nm) in combination with n-type TiO_2 to achieve RS performance and studied the interaction of Ag electrode with $CH_3NH_3PbBr_3$ under applied voltage, which affects the switching performance for long-term operation. The solution-processed $FTO/TiO_2/CH_3NH_3PbBr_3/Ag$ memristor shows a typical bipolar RS performance with low SET and RESET voltage at around 0.3 V and −1.4 V, respectively. However, a gradual decrease in reverse bias RESET voltage; hence, the reduction in low resistance state (LRS) current was observed with the increase in cycle number and finally stabilized at a lower LRS. X-ray photoelectron spectroscopy (XPS) in combination with energy dispersive X-ray spectroscopy (EDX) reveals that the formation of AgBr at the $CH_3NH_3PbBr_3/Ag$ interface in the presence of repetitive applied electric field changes the conduction process in LRS from filamentary conduction to Fowler-Nordheim (F-N) tunneling, which results in such performance degradation. Replacing the Ag electrode with Au, the $FTO/TiO_2/CH_3NH_3PbBr_3/Au$ device shows stable bipolar RS behavior with a completely interface-dominated conduction process. Our results

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demonstrate the fundamental limitation of the Ag electrode for $CH_3NH_3PbBr_3$-based memory devices and reveal that the bipolar switching mechanism in the proposed structure is highly dependent on the top electrode configuration.

II. EXPERIMENTAL SECTION

A. Materials and methods

Methyl ammonium bromide (CH_3NH_3Br), >98%, was purchased from TCI Chemicals. Other chemicals were >99% pure and purchased from Sigma-Aldrich. All the chemicals were used without any further purification. Firstly, before deposition, the FTO-coated glass substrates (sheet resistance: $30\,\Omega\,sq^{-1}$) were cleaned thoroughly using 2% Hellmanex solution, DI water, acetone, and methanol, followed by dry nitrogen blow. The oxide layer of TiO_2 was deposited following our previous reports [30,31]. In short, the TiO_2 layer was first deposited by spin-coating the 0.4 M acidic solution of titanium (IV) isopropoxide (dissolved in ethanol) at 2000 rpm for 30 s five times for a thickness of ∼200 nm with pre-annealing each layer at 200 °C. The films were finally annealed at 500 °C to obtain a crystalline n-type oxide layer.

A two-step spin coating method is used for ∼300 nm perovskite layer deposition. 1 M lead bromide ($PbBr_2$) solution prepared in N, N -dimethylformamide (with overnight stirring at 70 °C) was spin-coated on top of TiO_2 layer at 3000 rpm for 1 min in an Ar atmosphere glove box and annealed at 100 °C for 10 min; followed by dropping the CH_3NH_3Br solution (10 mg/ml in isopropyl alcohol with overnight stirring at 50 °C) with either substrates/solution at 60 °C, left for the 30 s to complete the $PbBr_2$ and CH_3NH_3Br reaction and spin-coated at 3000 rpm for 1 min to remove the extra solvent, and annealed at 110 °C for 30 min. [32]. Finally, 100 nm top electrodes of Ag or Au were thermally evaporated to complete the device structure.

B. Characterization

The X-ray diffraction (XRD) pattern for the as-deposited sample was characterized using Cu Kα radiation, $\lambda = 1.5406$ Å (Panalytical Empyrean) to study the structural composition, and the data were analyzed using X'Pert High Score software. The chemical composition and oxidation states of the main components (Pb, Br, and Ag) were studied using XPS (Model: Nexsa-ThermoFisher). The surface morphology and thickness of the thin films were examined by field emission scanning electron microscope (FESEM) (Model: JEOL JSM 7900F). All the *I-V* characteristics were measured with the help of a Keithley 4200 parameter analyzer by placing the samples in an enclosed probe station (Excel Instruments) in the dark (to avoid any indoor and outdoor light impacts) under ambient environmental conditions.

III. RESULTS AND DISCUSSIONS

The intense peaks at 2θ values of 15.1°, 21.3°, 30.3°, 33.8°, 43.3°, 46.1°, and 48.7° for the XRD pattern in Fig. 1(a) correspond to (100), (110), (200), (210), (220), (300), and (310) planes, respectively, confirm the formation of pure-phased $CH_3NH_3PbBr_3$ (crystallographic open database reference no. 00–067–0864) without any additional impurity. The top view FESEM image of the $CH_3NH_3PbBr_3$ thin film in Fig. 1(b) shows smooth surface morphology with uniform granular structure. XPS spectrum of the $CH_3NH_3PbBr_3$ film in Fig. 1(c) shows the presence of C, N, Pb, and Br, which defines the phase purity of the as-prepared sample. The narrow scan spectrum for the C 1s Fig. 1(d) was deconvoluted into three peaks, where the peak found at 284.8 eV is widely used for calibration and is known as adventitious carbon (C-C). The peak found at 286 eV can be attributed to methyl carbons in the $CH_3NH_3PbBr_3$ film and mainly corresponds to the C-N group. As the film thickness is in the nm range, a very low carbon peak at 288.4 eV occurs and can be assigned to $PbCO_3$, indicating the formation of a carbonate as an extra byproduct during the decomposition of $CH_3NH_3PbBr_3$ [33]. The high-resolution spectrum for the +2 oxidation state for Pb 4f shows peaks at 138.38 and 143.28 eV ascribed to $4f_{7/2}$ and $4f_{5/2}$ (See Supplemental Material at Fig. S1(a) [34]). Minor peaks on the right of the Pb^{2+} peaks belong to the Pb (0) artifacts observed during the XPS analysis under ultrahigh vacuum for lead halide perovskites [35]. Figure S1(b) of [34] shows a narrow scan peak at 68.28 and 69.38 eV associated with Br $3d_{5/2}$ and $3d_{3/2}$ and are assigned to the −1 oxidation state of Br in the pristine $CH_3NH_3PbBr_3$ film [36,37].

Figure 2(a) shows typical diode-like *I-V* characteristics for the FTO/TiO_2/$CH_3NH_3PbBr_3$/Ag device with circular Ag top electrode of diameter 600 μm in the pristine state under the applied bias of ±1 V conforming MHP/oxide heterostructure formation. Inset of Fig. 2(a) shows the cross-sectional FESEM image of the as-prepared device with a TiO_2 layer of thickness ∼200 nm and a perovskite layer of thickness ∼300 nm, which is much smaller than previous reports on $CH_3NH_3PbBr_3$ switching [26,28]. The n-type TiO_2 layer insertion with $CH_3NH_3PbBr_3$ results in asymmetric and typical analog switching behavior, as shown by the semi-log *I-V* plot in Fig. 2(b). The resistive switching behavior under an applied voltage of +2 to −3 V shows a gradual increase in current at +0.3 V, resulting in the SET process and self-compliance behavior due to Br-ions piling up near the top electrode under the influence of a positive field [38,39]. The device gradually gets RESET to the high resistance state (HRS) at around −1.4 V, showing bipolar RS behavior. Our heterojunction device's SET and RESET voltages are lower than the only $CH_3NH_3PbBr_3$-based memristor [27]. RS characteristics with individual layers of TiO_2 and $CH_3NH_3PbBr_3$ having Ag as the top electrode, along with their comparison with the heterostructure, are shown in Supplemental Material in Fig. S2 of [34]. The FTO/$CH_3NH_3PbBr_3$/Ag device gets SET around +1 V and RESET around −1.5 V with a meager ON-OFF ratio of <10 and low endurance stability that breaks easily even within the span of 30 cycles, persistently following the previous report [28]. On the other hand, the FTO/TiO_2/Ag device shows the SET and RESET voltage of around 1.2 V and around −2 V, respectively, with the ON-OFF ratio of ∼20, which deteriorates with the increase in switching cycles presented earlier in our previous report. [40] On comparison of RS characteristics with the heterostructure memory device, the bilayer structured device shows the superiority of the heterostructure with less SET-RESET voltage and improved endurance performance with a larger ON-OFF ratio.

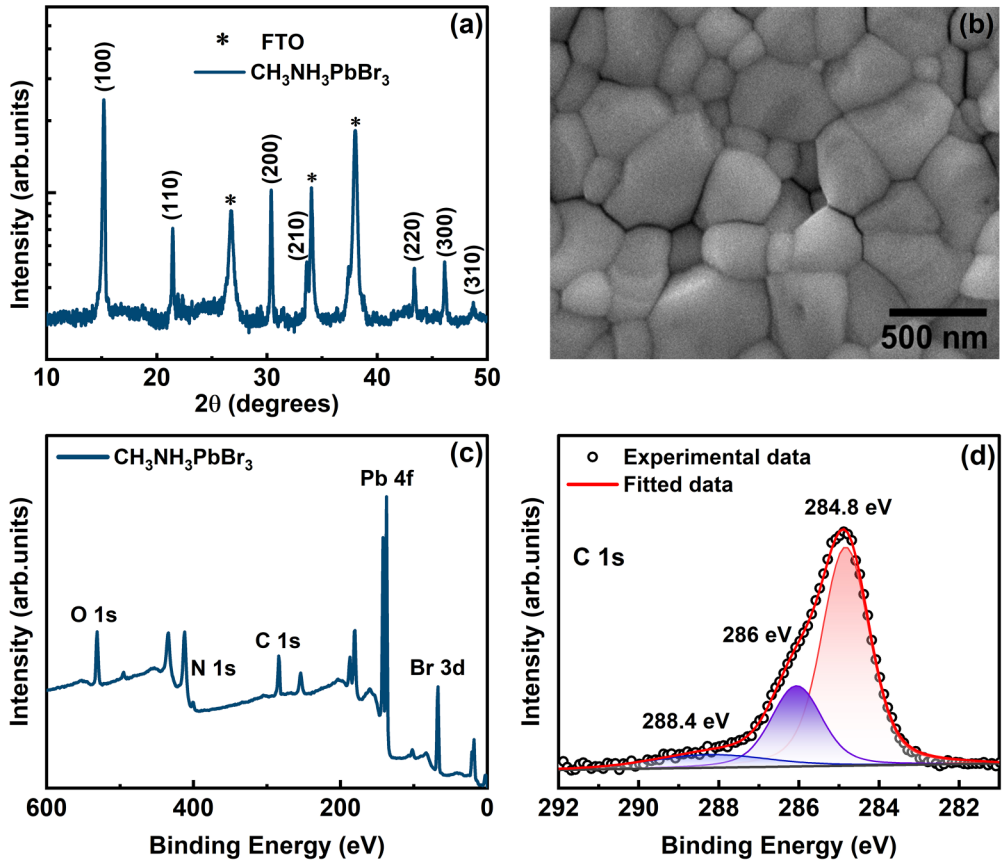


FIG. 1. (a) XRD pattern, (b) top-view FESEM image, (c) XPS survey scan spectrum of $CH_3NH_3PbBr_3$ film, and (d) high-resolution XPS spectrum of C 1s.

However, as we can see from Fig. 2(b), with an increase in the number of cycles in steps of 10 cycles, the switching behavior for the FTO/TiO_2/$CH_3NH_3PbBr_3$/Ag device shows a gradual decrease in the RESET voltage from −1.4 V to −0.3 V only after 40 cycles and the LRS current also decreases drastically resulting a decrease in ON/OFF ratio from ∼80 to a stable value of ∼10 just after 50 cycles when read at −0.5 V Fig. 2(c). The device shows a stable maximum ON-OFF ratio for up to 200 cycles when read at 0.1 V, as shown in the endurance plot (see Supplemental Material, Fig. S3 [34]). The retention data for the bilayer device up to 10^4 s shows a fluctuating decrease in HRS state and an increase in LRS when read at +0.2 V, resulting in a gradual reduction in ON/OFF ratio from 200 to ∼80 as shown in Fig. 2(d). A faster decay in the R_{ON}/R_{OFF} ratio in the retention curve was observed when read at −0.5 V (see Supplemental Material at Fig. S4 [34]). To check the performance repeatability of the solution-processed heterostructure, 25 randomly selected devices were tested for switching. The cumulative probability distribution curve shows uniformity in device-to-device variation data with a coefficient of variation $(\sigma/\mu) = 0.39$ for LRS and = 0.28 for HRS read at 0.1 V (see Supplemental Material in Fig. S5 [34]).

To understand the charge transport mechanism and associated device performance of FTO/TiO_2/$CH_3NH_3PbBr_3$/Ag memristor, we have fitted the *I-V* curves at different states as plotted in Figs. 3(a)–3(c). The double logarithmic *I-V* curve for the first cycle under positive bias shows a linear relation for LRS ($I \propto V$), confirming that the Ag ion-assisted filamentary conduction mechanism dominates after the SET process [41]. The device shows no significant change in LRS for RS performance by varying electrode size, further confirming filamentary conduction [42] (See Supplemental Material Fig. S6 of [34]). For positive bias, HRS follows the space-charge-limited-conduction (SCLC) mechanism Fig. 3(a) following Eq. (1) [40].

$$J_{\mathrm{SCLC}} = \frac{A\mu\varepsilon\varepsilon_0}{d^3}\exp\left(\frac{0.891}{kT}\sqrt{\frac{Vq^3}{d\pi\varepsilon\varepsilon_0}}\right), \qquad (1)$$

where J_{SCLC} is current density, A is a positive constant, k and T are Boltzmann constant, and temperature, q, ε_0, V denote electron charge, vacuum permittivity, and applied voltage, respectively, and ε, μ, d denote absolute permittivity, carrier mobility, and thickness of the dielectric layer, respectively. For low voltage region of 0–0.2 V exhibits ohmic conduction behavior with slope = 1 as the thermally generated charge carriers have significant relaxation time and do not travel across the entire layer; with further increase in voltage, the device gets SET at a voltage around 0.3 V with the slope ∼5.3 following trap-assisted SCLC mechanism [42,43]. From Fig. 3(b), the HRS state under reverse bias agrees with the hopping conduction mechanism, which assures the TiO_2/$CH_3NH_3PbBr_3$ interface plays a significant role in the RESET process

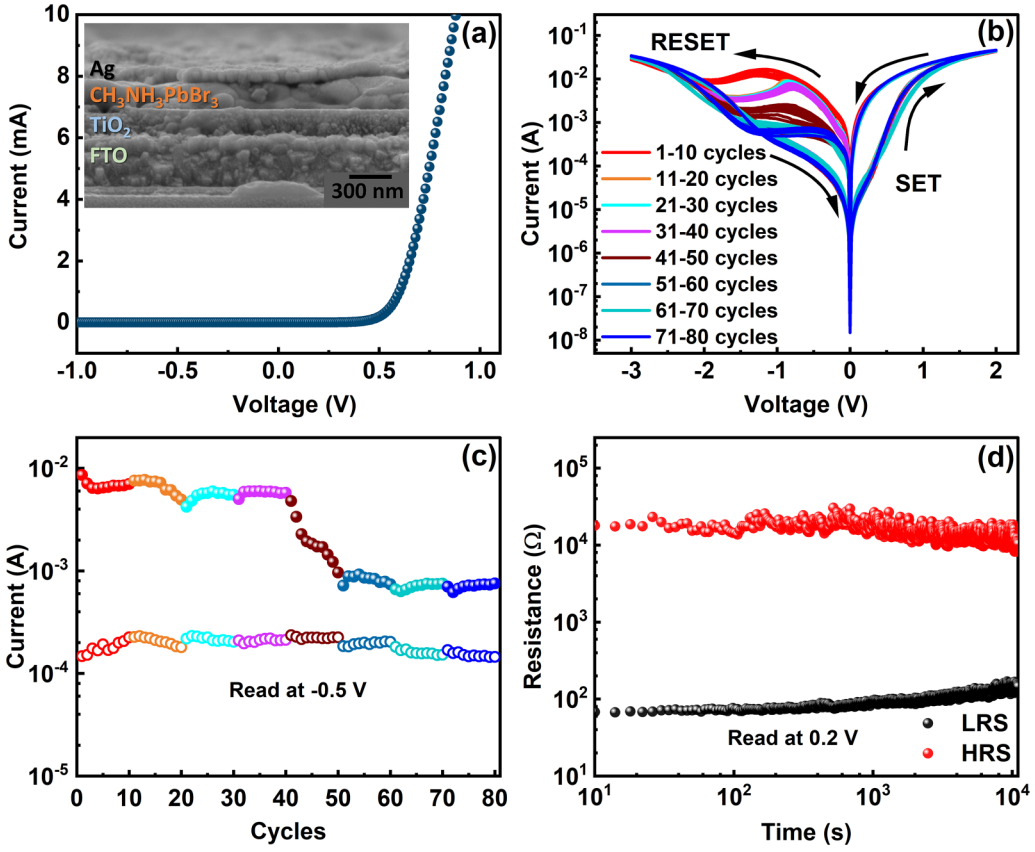


FIG. 2. (a) *I-V* characteristics for FTO/TiO_2/$CH_3NH_3PbBr_3$/Ag device (inset shows the cross-sectional FESEM image), (b) Typical RS characteristics, (c) semi-logarithmic endurance data plot read at −0.5 V and (d) retention data read at 0.2V for FTO/TiO_2/$CH_3NH_3PbBr_3$/Ag device.

following current density Eq. (2) [44].

$$J_{\text{Hopp}} = qnva \exp\left(\frac{qaE - \Phi_t}{k_B T}\right), \tag{2}$$

where q, E, k_B, and T stand for an elementary charge, applied electric field, Boltzmann constant, and temperature, respectively, and n is conduction band (CB) charge concentration, v is thermal vibration frequency at trap sites, a is hopping distance (distance between trap sites), and Φ_t denotes the energy difference between the CB and trap states.

The LRS under reverse bias follows ohmic conduction for the first cycle (inset of Fig. 3(b)), but the LRS current decreases with an increase in no. of cycles [Figs. 2(b) and 2(c)] also changes the charge conduction mechanism (see Supplemental Material Fig. S7 [34]. As shown in Fig. 3(c), the LRS at the reverse bias of the 60th cycle follows direct tunneling for low-voltage regions, and F-N tunneling behavior dominates for higher-voltage regions that can be expressed by Eqs. (3) and (4), respectively [45].

$$J_{\text{DT}} \propto E \exp\left(-\frac{4\pi(2m_e\Phi_0)^{1/2}}{h}\right), \tag{3}$$

$$J_{\text{FN}} = \frac{q^2}{8\pi h}\frac{E^2}{\Phi_0}\exp\left(-\frac{8\pi(2m_e q)^{1/2}}{3h}\frac{{\Phi_0}^{3/2}}{E}\right), \tag{4}$$

where q, m_e, and h denotes electronic charge, effective electron mass, and Planck's constant, respectively, Φ_0 is CB offset, and E is the applied electric field. Following the above equations, the plot of $\ln(I/V^2)$ vs V^{-1} shows a linear negative slope for F-N tunneling and an exponentially growing slope for direct tunneling [46,47]. This shows the complete change in the conduction process in the negative bias LRS from filamentary ohmic (electrode-dominated) to F-N tunneling (interface-dominated) after 60 cycles, whereas a minimal change in the forward bias conduction process was observed, and the slope changed to 1.35 (see Supplemental Material Fig. S8) [34]. We also observed a gradual change in the electrode color, from shinning Ag color toward red-brown with an increase in cycle number (Supplemental Material Fig. S9) [34], and a significant contrast of the Ag electrode color at the pristine state and after 80 cycles can be seen in the optical images given at the insets of Fig. 3(c).

We studied the changes in surface compositions and morphology of the Ag electrode by XPS and FESEM to understand the role of Ag-electrode in the performance degradation during cyclic operation of the FTO/TiO_2/$CH_3NH_3PbBr_3$/Ag memristor. XPS study shows a noticeable shift in the Ag 3d core level, as shown in Fig. 3(d). High-resolution XPS spectra for Ag^0 show metallic behavior with $3d_{5/2}$ level at 368.2 eV and $3d_{3/2}$ level at 374.2 eV, confirmed by the loss feature/plasmon peak at 371.8 eV. After switching, the 3d core level shows the shift of 0.4 eV with no loss feature, and peak

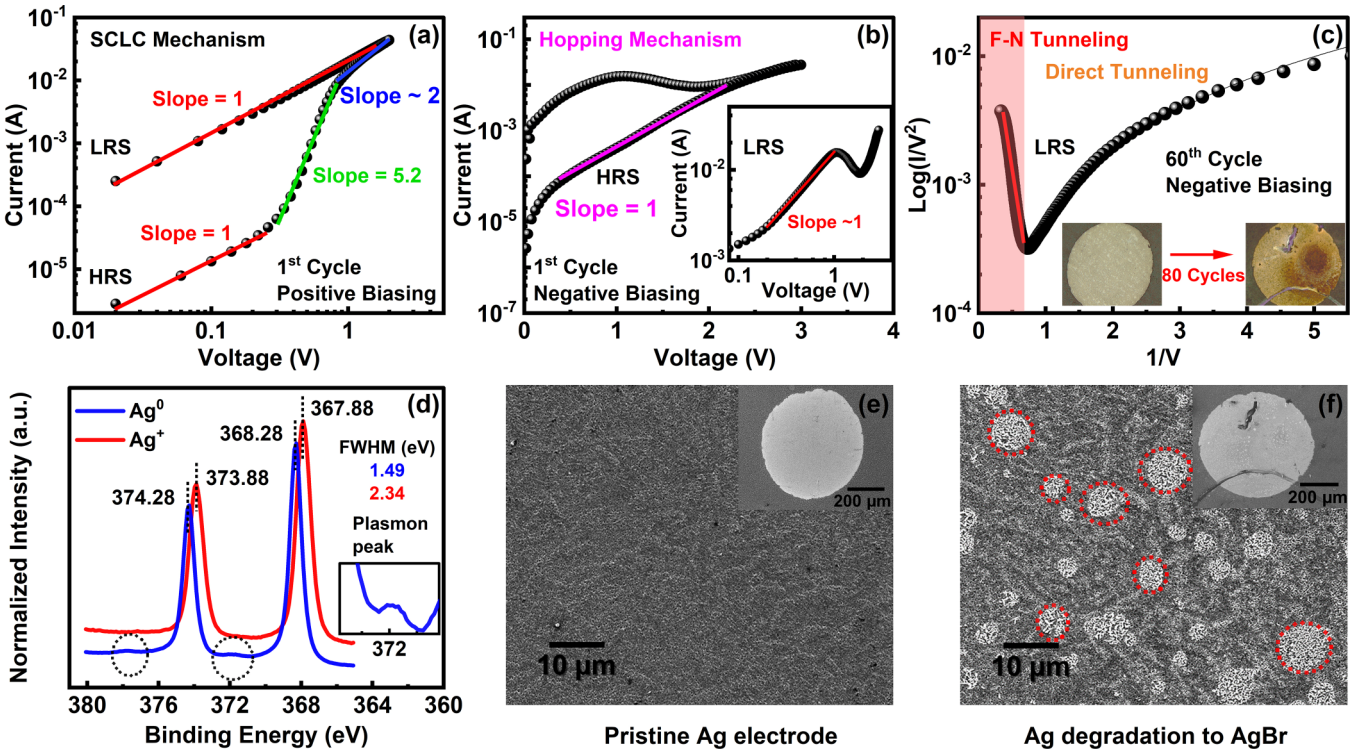


FIG. 3. (a) Double-logarithmic *I-V* plot for SET-process and (b) LogI vs V plot for RESET-process (first cycle). Inset shows LRS under negative bias. (c) F-N tunneling *I-V* fitting for LRS (insets show optical images for Ag electrode) for the 60th cycle. (d) High-resolution XPS spectra before and after RS performance. Top view FESEM image of Ag electrode (e) before and (f) after switching (Insets are the low-magnification FESEM).

broadening (increase in full-width at half maximum, FWHM value from 1.49 to 2.34 eV) confirms the Ag^+ oxidation state [48,49], showing the AgBr formation. The formation of AgBr can also be seen by the decrease in Ag atomic composition (from 24.19% to 8.18%) and the increase in Br atomic composition (18.90% to 30.42%) estimated from the XPS study (see Supplemental Material Table S1 of [34]) [50]. However, it is noteworthy to mention that we observed the Pb and Br signal in the pristine state XPS of the Ag electrode, which mainly appears due to the inhomogeneity of the solution-processed perovskite films, and the deposited silver does not form a continuous silver film instead tends to form particles on a bare perovskite surface, during evaporation [51]. Figures 3(e) and 3(f) show the top view FESEM image for the electrodes before and after switching, respectively, and the inset shows the low magnification images for the same. In contrast to the smooth surface of the pristine Ag electrode (3e), the formation of different sizes of patches (encircled in red in 3f) was observed after 80 cycles. The elemental composition data for the Ag electrode before and after switching using EDX (mapping shown in Supplemental Material Fig. S10) [34] shows the decrease in Ag concentration with the increase in Br ion concentration, as given in Supplemental Material Table S1 [34], confirming the conversion of the Ag electrode to AgBr after switching under the effect of an applied electric field. Furthermore, the EDX spectra of the individual patches show an enhanced Br concentration from 9% to 24%, and in addition, a change in Ag oxidation states confirms the formation of AgBr. A similar formation of AgI has been reported in the case of $CH_3NH_3PbI_3$ within the 7 days, with no prominent changes for bromide perovskite [51]. However, it is worth mentioning that AgBr formation was only observed after applying the voltage sweep cycles; no such color change was observed for the electrodes even after a month, showing that the AgBr formation is most likely due to the application of electric field (Supplemental Material Fig. S11) [34].

Based on the above observations, we schematically explained the charge transport phenomenon across the $FTO/TiO_2/CH_3NH_3PbBr_3/Ag$ device for different states in Fig. 4. Figure 4(a) shows the pristine state under zero applied voltage with oxygen vacancies in the TiO_2 layer, and the well-known Br vacancies (V_{Br}) and interstitials (I_{Br}), in the $CH_3NH_3PbBr_3$, which mainly participate in current conduction due to high defect concentration and low formation energy [52,53]. The I_{Br} moves toward the $CH_3NH_3PbBr_3/Ag$ interface under the applied positive field. The V_{Br} migration and the Ag ion migration from the top electrode result in conductive filament formation across the perovskite layer via grain boundaries (that provide dominant nonradiative recombination sites and act as the main channel for ion migration), resulting in electroforming free RS [54]. The oxygen vacancies (V_O) work as a virtual electrode for charge transport across the oxide layer for the SET process Fig. 4(b) [55]. The negative field applied at the top electrode forces oxygen vacancies toward the $TiO_2/CH_3NH_3PbBr_3$ interface, which bonds well with the I_{Br} present at the interface (migrated under the negative bias), resulting in a potential barrier across the junction. The hopping of charge carriers across the junction leads to filament rupture near the interface, which RESET the device as plotted in Fig. 4(c). The continuous ion migration

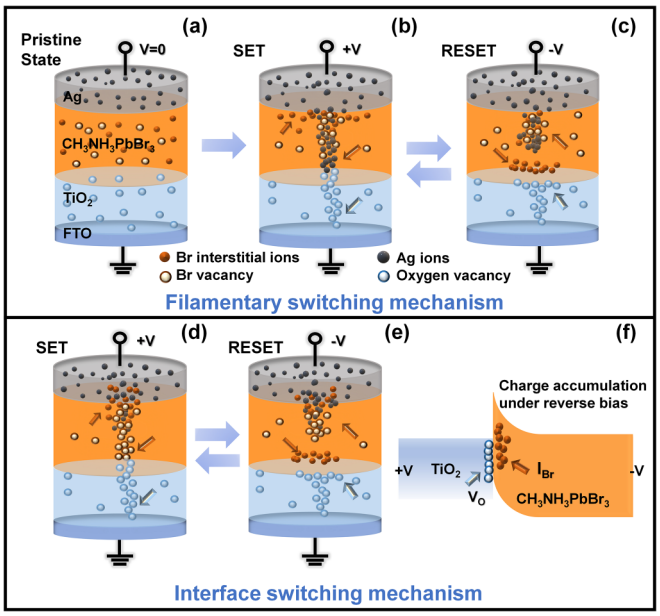


FIG. 4. Schematic illustration of charge conduction for $FTO/TiO_2/CH_3NH_3PbBr_3/Ag$ device (a) pristine state, (b) SET, and (c) RESET state for filamentary switching mechanism showing Ag-ions dominated conduction inside MHP layer. (d) SET and (e) RESET state by (f) charge accumulation showing defects-migration dominated interface switching mechanism.

toward both interfaces results in Ag ion transformation to AgBr near the $CH_3NH_3PbBr_3/Ag$ interface, which further prohibits the Ag ion migration through the perovskite layer [56]. As a result, the charge conduction process is dominated by the rest of the charge carriers that cross the barrier through F-N tunneling in the LRS. In this case, the defect migration (I_{Br} and V_{Br}) inside the MHP layer plays a vital role in charge transport under the effect of applied positive and negative fields for the SET and RESET process across the interface, as shown in Figs. 4(d) and 4(e). Figure 4(f) shows the oxygen vacancy and bromine interstitial ion migration-induced triangular interface barrier formation at reverse bias responsible for F-N tunneling. However, at forward bias, the $TiO_2/CH_3NH_3PbBr_3$ band alignment (Supplemental Material Fig. S12) [34] allows an easier charge transport across the interface despite AgBr formation, leading to minimum changes in the conduction process [57].

Ag electrode was replaced with an Au to overcome the problem of performance degradation. Au, being an inert electrode for perovskite halides, prevents its electrochemical reaction with the halide active layer [53]. An overlapped *I-V* loop for 80 cycles in Fig. 5(a) shows a stable RS performance of the $FTO/TiO_2/CH_3NH_3PbBr_3/Au$ device. In contrast to the decaying endurance curve at −0.5 V bias for the Ag electrode, no such decay was observed in the LRS and HRS current with stable retention for LRS read at −0.5 V, as given in (See Supplemental Material Fig. S13 [34]). The long-term endurance data for the Au-top electrode device Fig. 5(b) shows a good ON/OFF ratio of ∼300 for 1000 cycles without any significant changes, showing device stability in the long run. The retention data for up to 10^4 s in Fig. 5(c) shows stable

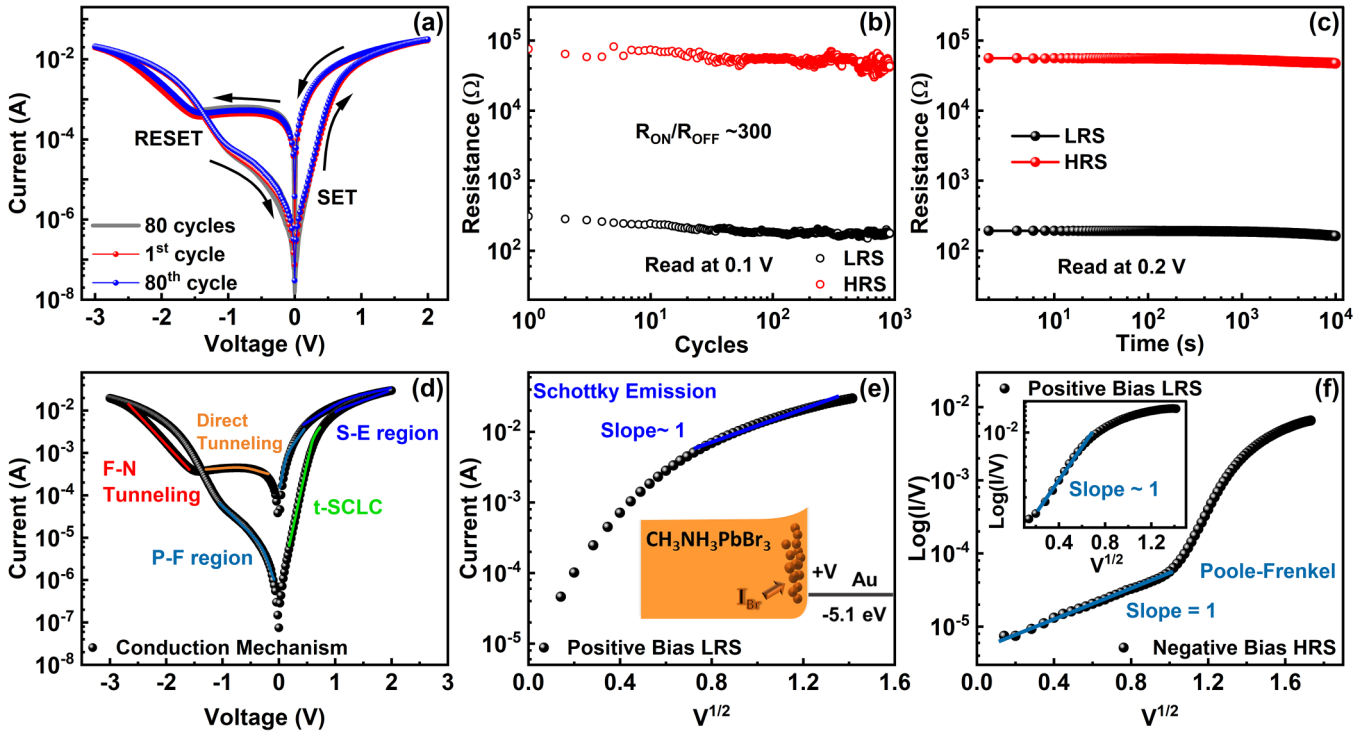


FIG. 5. (a) Typical RS characteristics, (b) endurance, and (c) retention data of $TiO_2/CH_3NH_3PbBr_3/Au$ device. (d) Semi-logarithmic *I-V* plot showing various conduction mechanisms. (e) Log *I* vs sqrt (*V*) plot for LRS. The inset shows a schematic illustration of the $CH_3NH_3PbBr_3/Au$ interface. (f) Log (*I*/*V*) vs sqrt (*V*) curve for low voltage HRS, and the inset shows the Poole-Frenkel curve at a low-voltage LRS under positive bias.

LRS and HRS, maintaining the R_{ON}/R_{OFF} ratio and assuring the nonvolatile nature of the device.

The curve-fitting analysis to study the switching mechanism for the Au-top electrode mainly follows the interface-dominated current conduction across the oxide/halide interface, as shown in Fig. 5(d). The device shows non-ohmic behavior for the LRS under both positive and negative bias, confirming the interface dominant conduction mechanism. Under low voltage region for positive bias, for HRS, the device follows the trap-assisted SCLC mechanism at the $TiO_2/CH_3NH_3PbBr_3$ interface for vacancies-oriented filament formation via grain boundaries, which is similar to Ag electrode-based device as plotted in Fig. S14a [34] in the Supplemental Material. Once the conducting filament formation is completed, in LRS, the current conduction is governed by the $CH_3NH_3PbBr_3$/Au interface at the higher positive voltage (Fig. S14b) [34]. However, unlike Ag, Au cannot be ionized to give an ohmic conduction; instead, it is followed by the Schottky-emission (S-E) at high positive bias Fig. 5(e) in the presence of a small Schottky barrier at the $CH_3NH_3PbBr_3$/Au interface (inset of Fig. 5(e)), as reported earlier [58]. The S-E mechanism (J_{SE}) follows Eq. (5) [40]:

$$J_{\mathrm{SE}} = A^* T^2 \exp\left[\frac{-q\left(\Phi_B - \sqrt{\frac{qE}{4\pi\varepsilon_r\varepsilon_0}}\right)}{kT}\right], \tag{5}$$

where T is the absolute temperature, A^*, k, q, and ε_r are the effective Richardson constant, Boltzmann constant, electronic charge, and relative permittivity, respectively; $q\Phi_B$ represents the Schottky barrier height. Under negative bias for the LRS state, the device follows the F-N tunneling for high voltage region and direct tunneling for low voltage region (Supplemental Material Fig. S15a [34] the linear fitting of $\ln(I/V^2)$ vs $\ln(1/V)$ supports direct tunneling). The charge accumulation near the interface under reverse bias resulted in a tunneling barrier for lower bias voltage ($V < -1.4$ V), which dominates the charge transfer with direct tunneling (Fig. S15b) [34,46,47]. However, at a higher voltage bias ($V > -1.4$ V), the sharper band bending resulted in a triangular tunneling barrier at the interface, resulting in the F-N tunneling (Fig. S15(c) [34]). During the RESET process for higher voltage regions of the HRS, the data crossover of the I-V curve at non-zero point arises due to dynamic transition from a fast capacitive to a slow inductive response in volatile perovskite-based memristors with inverse hysteresis loop [59,60]. The device exhibits the bulk-limited conduction mechanism for the low-voltage region, mainly Poole-Frenkel (P-F), as shown in Fig. 5(f), due to the trap-assisted charge transfer under the influence of a thermal or electric field. The current density (J_{PF}) for the P-F mechanism follows Eq. (6) [45,61]:

$$J_{\mathrm{PF}} = q\mu N_C E \exp\left[\frac{-q\left(\Phi_T - \sqrt{\frac{qE}{\pi\varepsilon}}\right)}{kT}\right], \tag{6}$$

where N_C is the density of states in the CB, μ is electronic drift mobility, and Φ_T is potential-well depth for the traps near CB. As the switching mechanism only depends on the movement of vacancies and interstitial ions near the $TiO_2/CH_3NH_3PbBr_3$ interface of the FTO/TiO_2/$CH_3NH_3PbBr_3$/Au-based memristor with almost no interference of the Au electrode, the device shows a stable switching performance.

IV. CONCLUSION

We designed a memristor device at the $TiO_2/CH_3NH_3PbBr_3$ interface, with a $CH_3NH_3PbBr_3$ thickness of only about 300 nm, and studied the role of Ag electrode in the performance of FTO/TiO_2/$CH_3NH_3PbBr_3$/Ag memristor device. Although the device shows a minimal SET voltage of 0.3 V with a good ON/OFF ratio over 100 for the first cycle, drastic deterioration of the switching behavior was observed after the 40th cycle, and electrode color changed to reddish-brown only after 80 cycles. XPS, FESEM studies combined with EDX analysis show that on the application of the repetitive voltage sweeps, the Ag electrode starts to interact with the Br ions, and the formation of AgBr leads to a change in the conduction mechanism in the LRS from conventional ohmic behavior via electrochemical metallization to a defect migration induced F-N tunneling mechanism. However, using Au as the top electrode, the FTO/TiO_2/$CH_3NH_3PbBr_3$/Au device shows a stable switching performance with an ON/OFF ratio over 300 and an entirely interface-dominated conduction mechanism. Our results show the fundamental degradation mechanism of Ag electrode in the $CH_3NH_3PbBr_3$-based memristor devices, which will help choose suitable electrodes for MHP-based optoelectronic devices.

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